Supplementary materials for article: Atlas of heteroatomic chemical affinity in bcc Fe

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Contents

S1 VASP parameters **2**

S2 Solubility in BCC Fe **3**
S2.1 Chemical potentials . . . 3
S2.2 Solution energies . . . 5
S2.3 Benchmarking . . . 7

S3 Substitution-Vacancy interaction **8**
S3.1 Binding energies . . . 8
S3.2 Benchmarking . . . 9
S3.3 Analysis . . . 9

S4 Substitution-Substitution solutes interaction in bcc Fe **12**
S4.1 Binding energies . . . 12
S4.2 Benchmarking . . . 15
S4.3 Analysis . . . 15

S5 Substitution-interstitial solutes interaction in bcc Fe **16**
S5.1 Binding energies . . . 16

S6 Machine learning **18**

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S1. VASP parameters

Calculations of the total energies and optimized geometries were performed in the framework of the density functional theory (DFT) [1, 2], projected augmented wave (PAW) method [3] for the ion-electron interaction as implemented in VASP code [4]. For Fe atom, eight electrons ($3d^7 4s^1$) are treated as valence. The calculations were spin polarized and the exchange-correlation functional used is the generalized gradient approximation (GGA) of Perdew-Burke-Ernzerhof [5]. The calculations were performed using an energy cutoff of 400 eV for the plane-wave basis set and converged with respect to the k-point integration. Periodic boundary conditions and the supercell approach was used for all calculations. Brillouin-zone (BZ) integrals were approximated using the special k-point sampling of Monkhorst and Pack [6]. The $3 \times 3 \times 3$ grid was used for 54-atom bcc supercell ($3 \times 3 \times 3$ two-atom cubic cells). For other structures k-point grid was adopted according to the size of BZ to obtain similar densities of k-points for all unit cells. The integration over the BZ was made using the Methfessel-Paxton smearing method [7] with 0.1 eV smearing width.

INCAR file containing the algorithm and parameter settings used for VASP calculations. Initial magnetic moments were set as 2.16 for Fe and 0.6 for other elements.

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KPOINTS file

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S2. Solubility in BCC Fe

S2.1. Chemical potentials

Table S1: Chemical potential references used for the calculation of solute formation and solution energies in bcc Fe. The elemental reference corresponds to the energy per atom of the most stable elemental phase (or molecular reference for diatomic gaseous species such as H_2, N_2, O_2, F_2, and Cl_2). For elements with available stable Fe-rich compounds, an alternative chemical potential is obtained from the corresponding Fe-rich phase. The Fe-rich chemical potential represents the maximum solute chemical potential allowed under equilibrium with bcc Fe and the listed binary phase. $\Delta\mu$ denotes the difference between the Fe-rich and elemental references, $\Delta\mu = \mu_{\text{Fe-rich}} - \mu_{\text{element}}$, and therefore quantifies the chemical potential reduction imposed by the competing Fe–X phase. MP-ID values are provided for elemental solid references when available.

Element	Reference	Phase / molecule	SG	MP-ID	μ (eV/atom)	$\Delta\mu$ (eV/atom)
H	Fe-rich	FeH	$Fm\bar{3}m$	mp-1009077	−3.352	−3.321
He	Atom in box	He	–	–	0.009	–
Li	Stable element	Li	$R\bar{3}m$	mp-1018134	−1.899	–
Be	Fe-rich	Be_2Fe	$P6_3/mmc$	mp-2225	−4.178	−0.421
B	Fe-rich	Fe_2B	$I4/mcm$	mp-1915	−7.635	−0.932
C	Stable element	C	$R\bar{3}m$	mp-569304	−9.233	–
N	Fe-rich	Fe_3N	$P6_322$	mp-1804	−8.617	−0.313
O	Fe-rich	FeO	$C2/m$	mp-1274279	−6.667	−1.739
F	Fe-rich	FeF_2	$P4_2/mnm$	mp-556911	−4.694	−2.920
Ne	Atom in box	Ne	–	–	−0.012	–
Na	Stable element	Na	$P6_3/mmc$	mp-10172	−1.313	–
Mg	Stable element	Mg	$R\bar{3}m$	mp-1094122	−1.512	–
Al	Fe-rich	$AlFe_3$	$Pm\bar{3}m$	mp-1183162	−4.539	−0.824
Si	Fe-rich	Fe_3Si	$Fm\bar{3}m$	mp-2199	−6.734	−1.314
P	Fe-rich	Fe_3P	$I\bar{4}$	mp-18708	−6.847	−1.475
S	Fe-rich	FeS	$P4/nmm$	mp-505531	−5.152	−1.025
Cl	Fe-rich	$FeCl_2$	$C2/m$	mp-571096	−2.911	−2.634
Ar	Atom in box	Ar	–	–	−0.025	–
K	Stable element	K	$Im\bar{3}m$	–	−1.013	–
Ca	Stable element	Ca	$P6_3/mmc$	mp-132	−1.921	–
Sc	Fe-rich	$ScFe_2$	$Fd\bar{3}m$	mp-540	−7.011	−0.811
Ti	Fe-rich	$TiFe_2$	$P6_3/mmc$	mp-2454	−8.570	−0.798
V	Fe-rich	VFe_3	$Fm\bar{3}m$	mp-866134	−9.417	−0.481
Cr	Stable element	Cr	$Im\bar{3}m$	mp-90	−9.470	–
Mn	Stable element	Mn	$I\bar{4}3m$	–	−8.983	–
Fe	Stable element	Fe	$Im\bar{3}m$	mp-13	−8.237	–
Co	Fe-rich	Fe_3Co	$P4/mmm$	mp-601820	−7.207	−0.170
Ni	Fe-rich	FeNi	$P4/mmm$	mp-2213	−5.589	−0.120
Cu	Stable element	Cu	$Fm\bar{3}m$	mp-30	−3.733	–
Zn	Fe-rich	Zn_9Fe_4	$I\bar{4}3m$	mp-1207450	−1.156	−0.056
Ga	Fe-rich	$GaFe_3$	$Pm\bar{3}m$	mp-19870	−3.461	−0.550
Ge	Fe-rich	Fe_3Ge	$Pm\bar{3}m$	mp-20344	−4.892	−0.411
As	Fe-rich	FeAs	$Pnma$	mp-427	−5.030	−0.349
Se	Fe-rich	FeSe	$P4/nmm$	mp-20311	−3.726	−0.228
Br	Fe-rich	$FeBr_2$	$C2/m$	mp-22880	−2.296	−0.662
Kr	Atom in box	Kr	–	–	−0.022	–
Rb	Stable element	Rb	$Im\bar{3}m$	–	−0.903	–
Sr	Stable element	Sr	$R\bar{3}m$	mp-1187073	−1.625	–
Y	Fe-rich	YFe_5	$P6/mmm$	mp-11385	−6.695	−0.265
Zr	Fe-rich	$ZrFe_2$	$Fd\bar{3}m$	mp-1718	−9.391	−0.872
Nb	Fe-rich	$NbFe_2$	$Fd\bar{3}m$	mp-1077790	−10.472	−0.371
Mo	Fe-rich	$Fe_{13}Mo_9$	$C2/m$	mp-1213182	−10.954	−0.027
Tc	Stable element	Tc	$P6_3/mmc$	mp-113	−10.365	–
Ru	Stable element	Ru	$P6_3/mmc$	mp-33	−9.259	–
Rh	Fe-rich	Fe_3Rh	$Fm\bar{3}m$	mp-1184364	−7.371	−0.091
Pd	Fe-rich	$FePd_3$	$Pm\bar{3}m$	mp-21845	−5.340	−0.112
Ag	Stable element	Ag	$R\bar{3}m$	mp-989737	−2.724	–
Cd	Stable element	Cd	$P6_3/mmc$	mp-94	−0.742	–
In	Stable element	In	$R\bar{3}m$	mp-1184502	−2.546	–

Sn	Fe-rich	$FeSn$	$P6/mmm$	mp-21260	−3.844	−0.026
Sb	Fe-rich	$FeSb_2$	$Pnnm$	mp-20714	−4.176	−0.028
Te	Fe-rich	$FeTe_2$	$Pnnm$	mp-19880	−3.407	−0.267
I	Fe-rich	FeI_2	$P\bar{3}m1$	mp-571122	−1.666	−0.148
Xe	Atom in box	Xe	–	–	−0.012	–
Cs	Stable element	Cs	$P6_3/mmc$	–	−0.868	–
Ba	Stable element	Ba	$Im\bar{3}m$	mp-122	−1.900	–
La	Stable element	La	$P6_3/mmc$	mp-26	−4.894	–
Ce	Fe-rich	$CeFe_2$	$Fd\bar{3}m$	mp-204	−6.152	−0.229
Pr	Stable element	Pr	$P6_3/mmc$	mp-38	−6.375	–
Nd	Stable element	Nd	$P6_3/mmc$	mp-123	−7.615	–
Pm	Stable element	Pm	$P6_3/mmc$	mp-867200	−9.206	–
Sm	Stable element	Sm	$R\bar{3}m$	mp-86	−11.117	–
Eu	Stable element	Eu	$Cmcm$	mp-1057315	−10.296	–
Gd	Fe-rich	Gd_2Fe_{17}	$P6_3/mmc$	mp-1196805	−14.058	+0.034
Dy	Fe-rich	$DyFe_3$	Cm	mp-1101819	−10.652	−2.268
Ho	Fe-rich	$HoFe_3$	$R\bar{3}m$	mp-30634	−8.197	+0.588
Er	Fe-rich	$ErFe_3$	$R\bar{3}m$	mp-569783	−7.741	−0.469
Tm	Fe-rich	$TmFe_3$	$R\bar{3}m$	mp-1103015	−6.835	−0.073
Lu	Fe-rich	$LuFe_2$	$Fd\bar{3}m$	mp-2463	−4.968	−0.444
Hf	Fe-rich	$HfFe_2$	$Fd\bar{3}m$	mp-333	−11.020	−1.059
Ta	Fe-rich	$TaFe_2$	$P6_3/mmc$	mp-568077	−12.432	−0.593
W	Fe-rich	Fe_2W	$P6_3/mmc$	mp-20868	−13.092	−0.083
Re	Fe-rich	Fe_2Re_3	$P4_132$	mp-639789	−12.452	−0.033
Os	Stable element	Os	$P6_3/mmc$	mp-49	−11.259	–
Ir	Fe-rich	$FeIr_3$	$I4/mmm$	mp-1184336	−8.932	−0.080
Pt	Fe-rich	$FePt$	$P4/mmm$	mp-2260	−6.584	−0.502
Au	Stable element	Au	$Fm\bar{3}m$	mp-81	−3.244	–
Hg	Stable element	Hg	$P6_3/mmc$	mp-1184554	−0.185	–
Tl	Stable element	Tl	$Im\bar{3}m$	mp-39	−2.222	–
Pb	Stable element	Pb	$Fm\bar{3}m$	mp-20483	−3.588	–
Bi	Stable element	Bi	$R\bar{3}m$	mp-23152	−3.886	–
Rn	Atom in box	Rn	–	–	−0.006	–
Ac	Stable element	Ac	$P6_3/mmc$	mp-862690	−4.071	–
Th	Fe-rich	$ThFe_5$	$P6/mmm$	mp-2689	−7.809	−0.408
Pa	Fe-rich	$PaFe_3$	$P6_3/mmc$	mp-1186420	−9.481	−0.072
U	Fe-rich	UFe_2	$Fd\bar{3}m$	mp-21050	−11.241	−0.107
Np	Stable element	Np	$Pnma$	–	−12.723	–
Pu	Stable element	Pu	$Fddd$	mp-571261	−13.974	–

S2.2. Solution energies

Table S2: Calculated solution energies, preferred incorporation sites, predicted equilibrium solubilities, and experimental solid solubilities of impurity elements in bcc Fe. Radii were taken from Mendeleev python library [8] (metallic_radius_c12 or, if missing, covalent_radius_pyykko)

Element	Radius	E_{oct}	E_{tet}	E_{sub}	Preferred site	T_{max} (K)	c_{max} (at.%)	c_{DFT} (at.%)
H	78	0.29	0.15	2.56	tet	1185	<0.1	23.0
Be	112	3.23	3.54	0.35	sub	1438	28.0	5.7
B	98	1.79	2.49	1.59	sub	1183	<0.1	10^{-5}
C	86	0.74	1.64	2.99	oct	996	<0.1	0.019
N	53	0.32	1.07	3.29	oct	863	0.39	1.4
Al	143	6.26	5.89	0.03	sub	1375	45.0	78.3
Si	138	5.10	4.92	0.07	sub	1473	3.2	59.5
P	128	3.79	3.77	0.35	sub	1321	4.8	4.5
S	127	3.42	3.50	1.20	sub	1639	<0.1	0.020
Sc	162	7.56	6.93	1.12	sub	1183	0.70	0.002
Ti	147	6.31	5.50	−0.01	sub	1562	10.0	> 100
V	134	5.41	4.88	−0.32	sub	1492	100.0	> 100
Cr	128	5.03	4.38	−0.26	sub	1103	100.0	> 100
Mn	127	5.11	4.23	0.10	sub	1011	3.0	31.0
Co	125	5.79	5.05	0.01	sub	1253	77.0	87.8
Ni	124	5.95	5.29	0.17	sub	618	5.5	4.5
Cu	128	6.45	5.91	0.70	sub	1758	6.7	1.0
Zn	134	6.32	5.93	0.31	sub	1053	42.0	3.2
Ga	140	6.06	5.77	0.11	sub	1310	36.0	37.1
Ge	144	5.68	5.38	−0.11	sub	1310	17.5	> 100
As	148	4.94	4.88	−0.07	sub	1113	10.0	> 100
Se	140	5.34	5.03	0.64	sub	1023	0.15	0.071
Y	180	9.71	8.78	2.27	sub	1530	<0.1	10^{-6}
Zr	160	8.68	8.44	1.09	sub	1608	<0.1	0.037
Nb	146	7.68	6.89	0.23	sub	1262	0.80	11.6
Mo	139	6.83	6.05	−0.05	sub	1724	24.4	> 100
Ru	134	7.09	5.81	0.31	sub	923	4.0	2.1
Rh	134	7.34	6.15	0.08	sub	878	19.0	34.3
Pd	137	8.13	7.04	0.50	sub	1088	3.5	0.5
In	158	9.00	8.20	1.24	sub	1221	0.28	0.001
Sn	163	8.73	7.82	0.66	sub	1183	9.2	0.2
Sb	166	8.53	7.55	0.48	sub	1185	4.5	0.9
Te	160	11.69	7.84	1.10	sub	1073	1.5	0.001
Gd	169	–	8.11	1.94	sub	1103	1.0	10^{-7}
Hf	159	8.45	7.58	0.95	sub	1663	0.70	0.1
Ta	146	7.65	–	0.10	sub	1220	0.80	38.3
W	139	7.32	6.46	−0.10	sub	1821	15.0	> 100
Re	137	7.11	6.06	−0.16	sub	1813	8.0	> 100
Os	135	7.45	6.09	0.15	sub	893	3.0	14.8
Ir	136	7.51	6.08	−0.16	sub	713	8.0	> 100
Au	144	9.21	7.93	0.92	sub	1161	1.0	0.010
Pt	139	8.32	6.99	0.16	sub	873	5.0	11.6
Np	171	–	4.88	0.04	sub	1000	<0.1	63.6
U	170	–	5.04	−0.12	sub	1353	<0.1	> 100

Table S3: The validation dataset comprising 44 elements with experimentally assessed equilibrium solubilities in bcc Fe. Elements are grouped according to chemical family and maximum equilibrium solid solubility. Experimental data were taken from assessed Fe–X phase diagrams [9].

Chemical group	c_{max} (at.%)	Elements
Interstitial	< 0.1	H, B, C
	0.1–1	N
3*d* transition metals	0.1–1	Sc
	1–10	Mn, Ni
	10–100	Ti, V, Cr, Co, Cu
4*d* transition metals	< 0.1	Y, Zr
	0.1–1	Nb
	1–10	Ru, Pd
	10–100	Mo, Rh
5*d* transition metals	0.1–1	Gd, Hf, Ta
	1–10	Os, Ir, Pt, Au
	10–100	W, Re
p-block	< 0.1	S
	0.1–1	Be, P, Se, In
	1–10	Si, As, Sn, Sb, Te
	10–100	Al, Zn, Ga, Ge
Actinides	< 0.1	U, Np

S2.3. Benchmarking

Table S4: Comparison of calculated solution energies with available first-principles literature data. Literature values are referenced to the stable elemental phase. Solution energies obtained using elemental chemical potentials are therefore listed together with the values obtained using Fe-rich reference chemical potentials employed in the present work whenever stable Fe–X compounds exist. A dash in the Fe-rich column indicates that no Fe-rich reference was available and the elemental reference was used.

Element	Site	Element ref.	Fe-rich ref.	Literature	Ref.
C	oct	0.74	–	0.72	[10]
N	oct	0.01	0.32	0.02	[11]
Sc	sub	0.31	1.12	0.24	[12]
Ti	sub	–0.81	–0.01	–0.88	[12]
V	sub	–0.80	–0.32	–0.72	[12]
Cr	sub	–0.26	–	–0.12	[12]
Mn	sub	0.10	–	0.16	[12]
Co	sub	–0.15	0.01	–0.31	[12]
Ni	sub	0.04	0.17	0.14	[12]
Cu	sub	0.70	–	0.76	[12]
Zn	sub	0.26	0.31	0.34	[12]

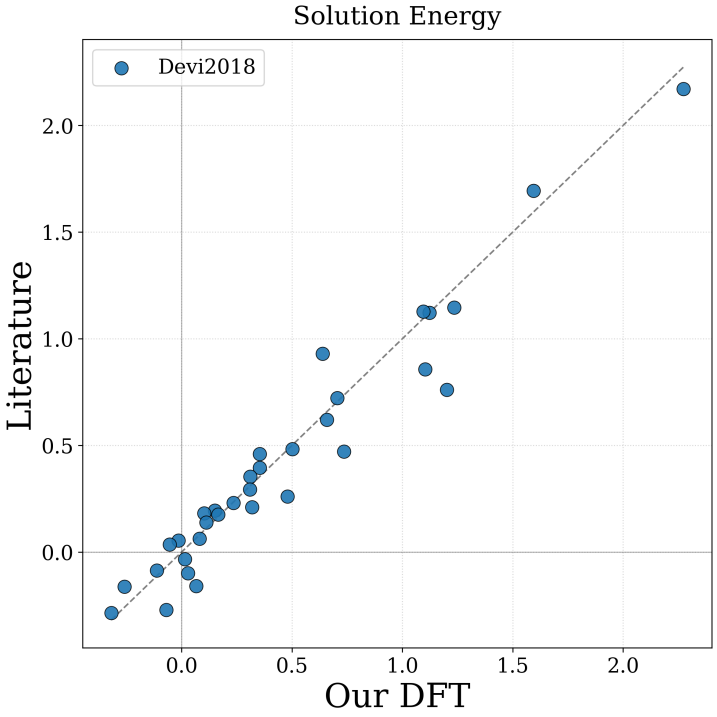


Figure S1: Comparison between our calculated and DFT values obtained by Devi et al. [13] for solution energies (in eV) in bcc Fe. Fe-rich reference chemical potentials were used in both cases.

S3. Substitution-Vacancy interaction

S3.1. Binding energies

Table S5: Binding energies E_b between solute atoms in substitutional position and vacancy in the bcc lattice of iron. nn - nearest neighbor.

Element	E_b (1nn), eV	E_b (2nn), eV	E_b (3nn), eV
Be	0.107	0.144	–0.008
B	0.172	0.323	–0.065
Al	0.349	0.020	0.062
Si	0.295	0.125	0.004
P	0.369	0.256	–0.021
S	0.573	0.377	0.004
Sc	0.709	–0.121	0.080
Ti	0.261	–0.160	0.016
V	0.056	–0.086	–0.008
Cr	0.050	0.005	–0.002
Mn	0.132	0.094	0.019
Fe	0.000	0.000	0.000
Co	–0.010	0.113	0.016
Ni	0.105	0.220	0.046
Cu	0.254	0.219	0.054
Zn	0.327	0.112	0.043
Ga	0.403	0.093	0.040
Ge	0.467	0.158	0.031
As	0.536	0.228	0.011
Se	0.675	0.308	0.014
Zr	0.745	–0.091	0.065
Nb	0.395	–0.142	0.053
Mo	0.170	–0.117	0.026
Ru	0.122	0.017	0.047
Rh	0.157	0.110	0.066
Pd	0.294	0.217	0.106
Sn	0.758	0.114	0.109
Sb	0.749	0.131	0.058
Te	0.801	0.196	0.031
Hf	0.646	–0.148	0.073
Ta	0.321	–0.180	0.040
W	0.129	–0.149	0.016
Re	0.077	–0.088	0.010
Os	0.091	–0.016	0.020
Ir	0.135	0.081	0.057
Pt	0.262	0.205	0.094
Au	0.457	0.297	0.138
U	1.248	0.229	0.073
Np	1.286	0.247	0.055

S3.2. Benchmarking

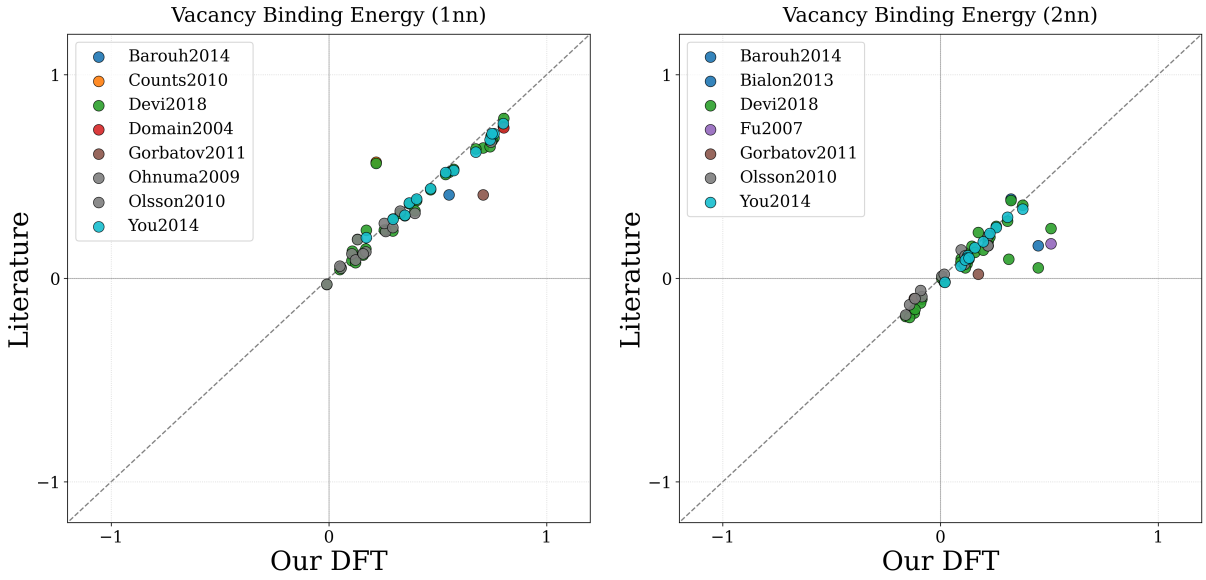


Figure S2: Comparison between our calculated and literature DFT values[13, 14, 15, 16, 17, 18, 19, 20] for 1-2nn solute-vacancy binding energies (in eV) in bcc Fe.

S3.3. Analysis

The calculated binding energies (E_b) between substitutional solutes and a vacancy across the first three coordination spheres (1nn, 2nn, and 3nn) are compiled in Table S5. As systematically illustrated in Fig. S3, the strongest attractive interactions in the first coordination sphere (1nn) are strictly governed by geometric and thermodynamic boundary conditions. Specifically, elements characterized by low intrinsic bulk solubility and atomic radii significantly exceeding that of Fe (e.g., S, Sc, Zr, Sn, Sb, Te, and actinide elements) exhibit the most pronounced vacancy-trapping capabilities.

This prominent trend reflects the substantial elastic strain relief achieved via the partial compensation of local lattice distortions; an oversized solute atom adjacent to a vacancy effectively relaxes into the vacant site, drastically lowering the total strain energy of the bcc Fe matrix. This structural argument is quantitatively supported by the strong positive correlation ($R^2 = 0.645$, $r = 0.803$) between the maximum solute–vacancy binding energy ($E_b^{X\text{-}Vac}$) and the relative atomic size mismatch, as demonstrated in Fig. S3a. Furthermore, Fig. S3b reveals a clear inverse linear relationship between $E_b^{X\text{-}Vac}$ and the logarithm of experimental solubility ($\log_{10}(C_{\text{exp}})$), reinforcing the concept that vacancy trapping acts as a fundamental atomic-scale stabilizer for chemically incompatible solutes within the ferrite matrix.

At larger defect separations, the elastic stress fields decay, causing the binding energies to generally decrease and approach weakly attractive or repulsive baseline values. However, the directional nature of chemical bonding and localized magnetic perturbations in the ferromagnetic (FM) host induce several critical non-monotonic anomalies:

- **Actinide Vacancy-Trapping Echelon:** The uppermost boundary of the 1nn dataset is defined by the actinide elements uranium and neptunium, which possess exceptionally high binding energies of 1.248 eV and 1.286 eV, respectively. In the context of nuclear materials science, these extreme affinities indicate that heavy radiation-induced solutes will act as immovable sinks for vacancies, driving the formation of complex actinide-vacancy clusters that can accelerate radiation embrittlement or void swelling.

- **The Boron 2nn Anomaly:** A highly anomalous behavior is observed for boron forced into a substitutional site. While the 1nn binding energy is modest (0.172 eV), the binding energy spikes significantly in the second coordination sphere to $E_b(2\text{nn}) = 0.323$ eV. This indicates that a substitutional B atom preferentially sits at the 2nn position relative to a vacancy. This counter-intuitive trend is rooted in the electronic structure of the FM bcc Fe lattice, where the directional covalent component of the *p-d* hybridization between boron and Fe atoms is energetically optimized when the vacancy does not directly interrupt the immediate 1nn bonding envelope.

- **Anisotropic Repulsion of d-Transition Metals:** A distinct group of hcp (Sc, Ti, Zr, Hf) and bcc (V, Nb, Mo, Ta, W, Re) refractory transition metals exhibits a clear repulsive interaction with vacancies exclusively at the 2nn distance (e.g., $E_b(\text{Sc, 2nn}) = -0.121$ eV, $E_b(\text{Ta, 2nn}) = -0.180$ eV). This systematic suppression

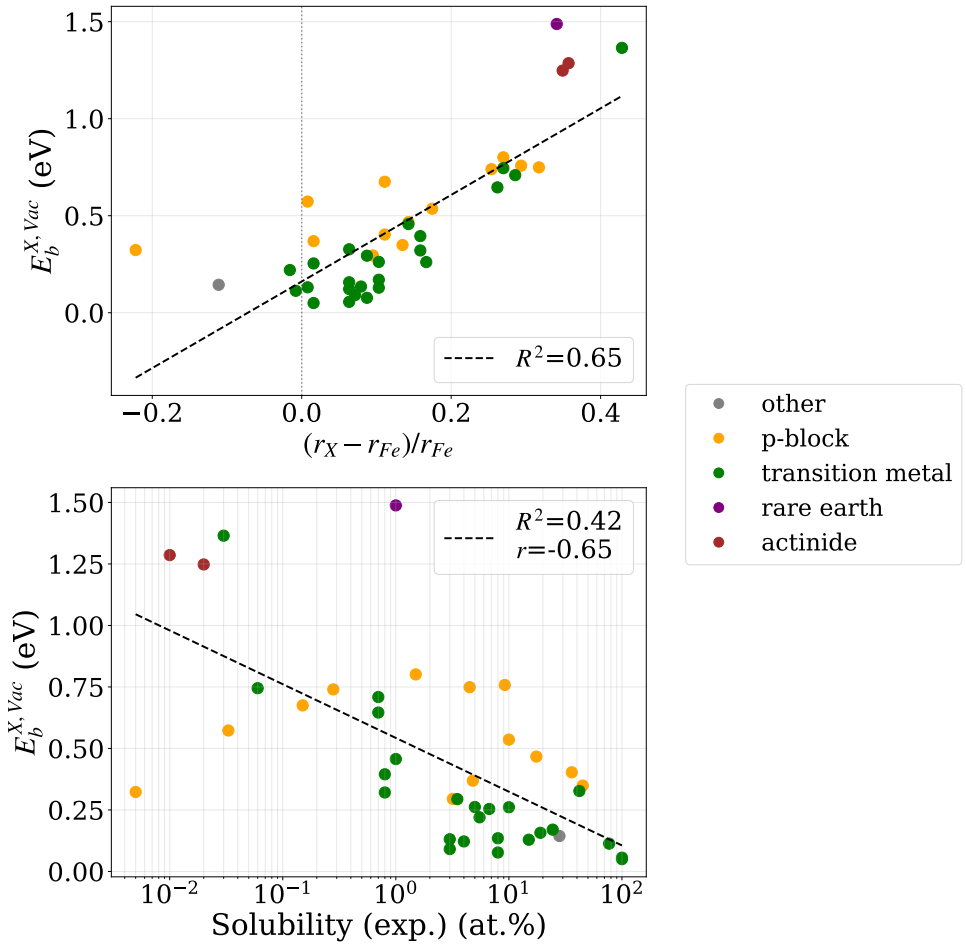


Figure S3: Binding energies $E_b^{X,\mathrm{Vac}}$ between a substitutional atom X and a vacancy in bcc iron, calculated using density functional theory (DFT), as a function of: (top) the relative difference in atomic radii $(r_X - r_{\mathrm{Fe}})/r_{\mathrm{Fe}}$, and (bottom) the experimental solubility of elements (at.%). The dashed lines represent linear regression fits with corresponding coefficients of determination R^2 and correlation r. Color coding indicates different groups of elements in the periodic table.

of 2nn affinity is a hallmark of early d-block elements in bcc iron. It stems from the spatial orientation of the d-orbitals (t_{2g} vs. e_g) and the corresponding local redistribution of spin polarization around both the vacancy and the open-shell solute, which creates a localized quantum-mechanical energy barrier at the 2nn position despite an attractive 1nn envelope.

In contrast, p-block elements like Sn (0.758 eV), Sb (0.749 eV), and Te (0.801 eV) maintain a strictly monotonic decay from 1nn to 3nn. Their interactions are predominantly electrostatic and elastic in nature, with their closed d-shells minimizing anisotropic magnetic scattering in the iron matrix, thereby allowing size-mismatch effects to dominate the trapping energetics.

Figure S4 presents the calculated effective enhancement factor of the equilibrium vacancy concentration, η, induced by solute–vacancy binding in bcc Fe at $T = 1200$ K. The actinides (U and Np) demonstrate the most pronounced effect, increasing the vacancy concentration by more than five orders of magnitude. This strong stabilization is driven by their low effective energy, $E_{\mathrm{eff}} = E_{\mathrm{sol}} - E_b^{\mathrm{max}}$. In contrast, p-block elements (such as As and Ge) and transition metals exert a significantly weaker impact, with η values dropping below 10^3 and 10^2, respectively.

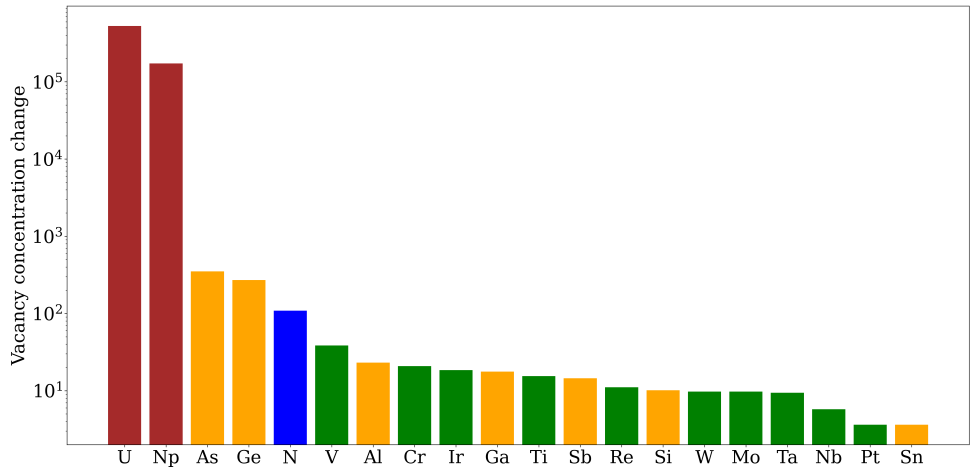


Figure S4: Effective enhancement of the equilibrium vacancy concentration in bcc Fe due to solute–vacancy binding, estimated using $\eta = 1 + \exp[-(E_{\mathrm{sol}} - E_b)/(k_{\mathrm{B}}T)]$ at $T = 1200$ K. The bars are colored according to the chemical group of the solute elements. The numbers above the bars indicate the effective energy $E_{\mathrm{eff}} = E_{\mathrm{sol}} - E_b^{\mathrm{max}}$, which determines the magnitude of vacancy stabilization.

S4. Substitution-Substitution solutes interaction in bcc Fe

S4.1. Binding energies

Table S6: Binding energies between solute atoms in substitutional positions in bcc Fe. S-, B- and Cu- containing pairs are highlighted by yellow, blue and red, respectively

Elements	E_b, eV	Elements	E_b, eV	Elements	E_b, eV	Elements	E_b, eV	Elements	E_b, eV
S–Gd	0.66	Sc–Ru	0.04	Mn–Pd	−0.08	Ga–Os	−0.21	Mo–Sb	−0.37
S–Y	0.64	Al–Co	0.04	Ni–Pt	−0.08	Al–Mo	−0.21	Ti–Ta	−0.37
B–Gd	0.62	B–Ru	0.04	Cr–Rh	−0.08	Ga–Zr	−0.21	Sc–W	−0.37
B–Y	0.62	Cu–Te	0.04	B–Cr	−0.08	Ga–Ge	−0.21	Mo–Te	−0.37
S–Sc	0.44	Co–Sb	0.04	Cr–Ir	−0.08	Se–Np	−0.21	Sc–U	−0.37
S–S	0.41	S–V	0.03	Ni–Rh	−0.08	P–Ga	−0.21	Mo–Au	−0.37
B–Sc	0.41	Co–Hf	0.03	Ru–Sn	−0.08	Al–Re	−0.21	Y–Sn	−0.37
B–Np	0.38	In–Au	0.03	As–Pt	−0.08	Ge–Mo	−0.21	Ti–Np	−0.37
S–Zr	0.36	Ti–Pd	0.03	P–Mo	−0.08	Ge–Nb	−0.21	Pd–Os	−0.37
B–Zr	0.35	Cu–Hf	0.03	Co–Ir	−0.08	Ga–Se	−0.21	Zn–U	−0.37
S–Np	0.35	Zn–Rh	0.03	Ru–Ru	−0.08	Al–Y	−0.21	Mo–Hf	−0.38
B–In	0.31	Si–Ir	0.03	Cr–Co	−0.08	Si–In	−0.21	Zr–Te	−0.38
B–Hf	0.31	Y–Pt	0.03	Ga–In	−0.09	Ga–Hf	−0.21	W–Re	−0.38
Cu–In	0.28	Cu–Zr	0.03	Be–Mo	−0.09	Zn–Ta	−0.21	Y–Sb	−0.38
S–Cu	0.27	Al–Ru	0.03	Sc–Os	−0.09	Mn–Ta	−0.21	Ta–Re	−0.38
S–Hf	0.27	B–B	0.03	V–Pd	−0.09	Al–Sn	−0.21	In–Sn	−0.38
S–Se	0.25	S–Ta	0.03	V–Ni	−0.09	Ta–Os	−0.22	Hf–Re	−0.38
S–U	0.25	Rh–Sb	0.03	Mn–Rh	−0.09	Ni–Re	−0.22	Mo–In	−0.38
B–U	0.24	Si–Co	0.03	Zn–Ru	−0.09	Rh–Au	−0.22	Ir–Au	−0.38
Cu–Sn	0.24	Ni–Zr	0.03	Ti–Os	−0.09	Al–As	−0.22	Nb–Re	−0.39
Al–Cu	0.24	Al–S	0.03	Ru–Sb	−0.09	Sc–Sc	−0.22	Nb–Te	−0.39
Be–In	0.23	Co–In	0.03	As–Zr	−0.09	V–Te	−0.22	Mo–W	−0.39
B–Sn	0.23	Zr–Rh	0.03	Be–Cr	−0.09	Re–Ir	−0.22	Nb–Mo	−0.39
S–Pd	0.22	S–Pt	0.03	Nb–Ru	−0.09	Cr–Au	−0.22	Y–Mo	−0.40
Se–Y	0.22	Pd–Hf	0.02	Ga–Ga	−0.09	Mo–Pd	−0.22	In–Hf	−0.40
Al–Pd	0.21	Zn–Pt	0.02	Mn–Ir	−0.09	Mn–Np	−0.22	Mo–Ta	−0.40
Cu–Ga	0.21	Co–Zr	0.02	Co–Ru	−0.09	Mn–Nb	−0.22	Zr–Mo	−0.40
Se–Gd	0.21	Ni–Au	0.02	Si–S	−0.09	Ti–In	−0.22	Ge–Sn	−0.40
Be–Au	0.21	Co–Cu	0.02	Si–Os	−0.09	V–Mn	−0.23	Zr–In	−0.40
Be–Y	0.20	Ti–Se	0.02	Ti–Zn	−0.09	Mo–Os	−0.23	Ga–Sb	−0.41
B–S	0.20	Sc–Zn	0.02	Nb–Ir	−0.09	Ge–In	−0.23	Sc–Ta	−0.41
Sc–Se	0.20	Ti–Ni	0.02	Mo–Ru	−0.10	Mo–Pt	−0.23	Sn–Gd	−0.41
Cu–Zn	0.20	Be–Se	0.02	Cu–Mo	−0.10	V–Sb	−0.23	Nb–Sb	−0.41
Cu–Cu	0.20	Mn–Sn	0.02	Pd–Au	−0.10	As–Os	−0.23	Nb–Sn	−0.41
Cu–Ge	0.20	Co–As	0.02	Al–In	−0.10	Sc–V	−0.23	Te–Hf	−0.41
Be–Gd	0.20	Ti–Co	0.02	Mn–Y	−0.10	Mn–W	−0.24	Sb–Gd	−0.41
B–Cu	0.19	S–Co	0.02	Al–Cr	−0.10	Mn–U	−0.24	Ge–As	−0.42
Cu–Au	0.19	S–Sb	0.02	Cr–Ge	−0.10	Pd–Np	−0.24	P–Sb	−0.42
Be–Cu	0.19	Co–Ge	0.01	Zn–As	−0.10	Cr–Gd	−0.24	Nb–In	−0.42
S–Au	0.18	P–Co	0.01	Be–V	−0.10	Nb–Os	−0.24	Si–As	−0.42
P–Y	0.18	Co–Se	0.01	Ru–Te	−0.10	V–Sn	−0.24	V–Np	−0.42
Ni–Sn	0.18	Mn–Ga	0.01	Mn–Os	−0.10	Cr–Mn	−0.24	Te–Os	−0.42
S–Mn	0.18	Be–Si	0.01	As–Ir	−0.10	Os–Os	−0.24	Ti–U	−0.42
Sc–Pd	0.18	Co–Te	0.01	Mn–Co	−0.10	Ni–Os	−0.24	Zr–Sb	−0.42
Cu–Y	0.18	S–Ge	0.01	Rh–Np	−0.10	Rh–Pt	−0.24	P–Te	−0.42
Si–Cu	0.18	Co–Zn	0.01	Pd–Pd	−0.10	Ti–V	−0.24	Zr–Sn	−0.43
Al–Au	0.18	Ga–Pt	0.01	Ru–W	−0.10	Ga–Nb	−0.24	Zr–Re	−0.43
P–Gd	0.17	B–Pt	0.01	Co–Re	−0.11	Ti–Sb	−0.24	Se–Te	−0.43
S–In	0.17	Rh–In	0.01	Cu–Os	−0.11	Si–Ta	−0.24	Y–Re	−0.43
Si–Pd	0.17	Ti–Ru	0.01	Ge–Y	−0.11	Ga–Sn	−0.24	Os–Pt	−0.43
Cu–Gd	0.17	Be–Ru	0.01	Rh–U	−0.11	Ti–Re	−0.24	Al–Te	−0.43
Al–Ni	0.17	Ge–Au	0.01	Rh–W	−0.11	Ga–Mo	−0.24	W–Au	−0.43

S–Ni	0.17	P–Au	0.01	Au–Au	−0.11	V–V	−0.24	Os–Np	−0.43
B–Zn	0.16	As–Y	0.01	Mo–Rh	−0.11	Pd–W	−0.24	As–As	−0.43
P–Cu	0.16	S–Mo	0.01	V–As	−0.12	W–Os	−0.25	As–Np	−0.43
B–Se	0.16	Be–Mn	0.00	Si–Y	−0.12	Ti–Sn	−0.25	Nb–Nb	−0.43
Ni–In	0.16	P–Hf	0.00	B–Re	−0.12	Rh–Ir	−0.25	W–W	−0.44
Be–Sn	0.16	As–Ru	0.00	Cr–Zn	−0.12	Si–W	−0.25	Zr–Gd	−0.44
P–Pd	0.16	B–Ir	0.00	Zn–Hf	−0.12	Rh–Os	−0.25	Sn–Re	−0.44
Ni–Y	0.15	Co–Ga	0.00	Se–Pt	−0.12	As–Ta	−0.25	Sn–Hf	−0.44
S–Ti	0.15	Mn–In	0.00	Sc–In	−0.12	Te–Ir	−0.25	Se–Sb	−0.45
Cu–Pd	0.15	Be–As	−0.00	Ga–Y	−0.12	Y–Os	−0.25	Cr–Np	−0.45
Be–Pd	0.15	Se–Ru	−0.00	Hf–Au	−0.12	Pd–U	−0.25	Sb–Hf	−0.45
P–Sc	0.15	B–Ni	−0.00	Pd–Ta	−0.12	Zn–Te	−0.25	V–U	−0.46
Sc–Ni	0.15	Sc–Ge	−0.01	Be–Re	−0.12	Cr–Hf	−0.25	Zr–Nb	−0.46
Sc–Rh	0.15	Rh–Te	−0.01	Zn–Zr	−0.12	Ir–Np	−0.25	Nb–Hf	−0.46
B–Nb	0.15	Hf–Pt	−0.01	Ti–Ge	−0.12	V–In	−0.25	Nb–W	−0.46
Cu–As	0.15	Zn–Y	−0.01	Be–W	−0.12	Te–Pt	−0.25	Nb–Gd	−0.46
B–Au	0.14	B–Mo	−0.01	P–Ta	−0.13	V–Cr	−0.25	In–Re	−0.46
Sc–Pt	0.14	Be–Co	−0.01	Ta–Pt	−0.13	Ru–Pd	−0.25	Gd–Re	−0.46
Cu–Se	0.14	P–Ti	−0.01	Al–Sc	−0.13	Pd–Gd	−0.25	Y–Nb	−0.46
Ge–Pd	0.14	Y–Au	−0.01	P–Os	−0.13	Sc–Y	−0.26	Ta–W	−0.47
Si–Ni	0.14	Ti–Cu	−0.01	V–Os	−0.13	W–Pt	−0.26	Os–U	−0.47
B–Al	0.14	Se–Hf	−0.01	Ru–Re	−0.13	Zr–Os	−0.26	As–Sn	−0.47
S–Nb	0.14	Pd–Te	−0.01	S–W	−0.13	Al–Nb	−0.26	Si–Np	−0.47
Al–Pt	0.14	Ge–Ru	−0.01	Sn–Ir	−0.13	V–Re	−0.26	Y–Zr	−0.47
Ga–Pd	0.14	Be–Te	−0.01	Be–Np	−0.13	P–U	−0.26	As–U	−0.47
Ni–Ge	0.14	Hf–Ir	−0.01	Cu–W	−0.13	Sc–Re	−0.26	Nb–Ta	−0.47
Be–Zn	0.14	Co–Ta	−0.01	Co–Os	−0.13	Re–Os	−0.27	Al–Np	−0.48
Ni–Sb	0.13	V–Ir	−0.02	Nb–Pd	−0.13	Mn–Pt	−0.27	Hf–W	−0.48
Pd–Sn	0.13	V–Rh	−0.02	As–Hf	−0.13	Zn–W	−0.27	Cr–U	−0.48
P–Ni	0.13	Cu–Ir	−0.02	Ti–Mn	−0.13	As–Se	−0.27	Y–Np	−0.48
S–Zn	0.13	Ge–Pt	−0.02	Ru–Np	−0.13	Zn–Re	−0.27	Gd–Hf	−0.48
Al–Rh	0.13	P–Pt	−0.02	Si–Ti	−0.13	Al–Zr	−0.27	Ge–Gd	−0.48
Ni–Cu	0.13	Zn–Sn	−0.02	Mo–Ir	−0.14	Ti–Mo	−0.27	Ga–Gd	−0.48
Sc–Cu	0.13	Zr–Pd	−0.02	Sc–Te	−0.14	Cr–Re	−0.27	Ga–Np	−0.48
Si–Rh	0.13	Ni–Pd	−0.02	Cu–Re	−0.14	Sc–Mo	−0.27	Ga–Te	−0.49
Pd–In	0.13	Mn–Zn	−0.02	Ni–Ir	−0.14	Ir–U	−0.27	Sb–Re	−0.49
Zn–Zn	0.13	Rh–Ta	−0.02	Cr–Ga	−0.14	Se–Os	−0.27	Si–U	−0.49
Cu–Sb	0.13	P–Cr	−0.02	W–Ir	−0.14	Co–Gd	−0.27	In–Ta	−0.49
Ni–As	0.13	Co–Nb	−0.02	Cr–Ni	−0.14	Cr–Zr	−0.27	Al–U	−0.49
Sc–Au	0.13	Zn–Ge	−0.03	Ni–Mo	−0.14	Al–Hf	−0.28	Zr–Zr	−0.50
Be–Sc	0.13	B–Si	−0.03	Ru–In	−0.14	Nb–Au	−0.28	Au–Np	−0.50
B–Sb	0.13	As–Gd	−0.03	Si–Gd	−0.14	Sc–Ti	−0.28	Sn–W	−0.50
B–Ga	0.13	P–Ir	−0.03	Al–Si	−0.14	Pd–Ir	−0.28	Zr–W	−0.50
Ni–Gd	0.13	Co–Co	−0.03	Si–V	−0.14	Se–U	−0.28	Te–Ta	−0.50
Y–Pd	0.12	Be–Ti	−0.03	Ru–Os	−0.14	Ti–Ti	−0.28	Sn–Ta	−0.50
Ni–Se	0.12	Gd–Pt	−0.03	Mn–Re	−0.14	Au–Pt	−0.28	Ga–U	−0.50
B–Ti	0.12	Zn–Gd	−0.03	Ti–Ga	−0.14	Os–Ir	−0.28	Y–Hf	−0.50
P–Rh	0.12	Ni–Ni	−0.03	V–Zn	−0.14	V–Mo	−0.28	Ta–Ta	−0.51
S–Rh	0.12	Be–Nb	−0.03	Cr–Pd	−0.14	Ga–As	−0.28	Zr–Np	−0.51
B–Pd	0.12	Mn–Sb	−0.03	Se–Mo	−0.14	P–Sn	−0.29	Zr–Hf	−0.51
Be–Pt	0.12	Be–P	−0.03	Cr–Pt	−0.14	Cr–Mo	−0.29	Si–Sb	−0.51
As–Pd	0.12	Cr–Cu	−0.03	Mn–Mn	−0.15	Ta–Au	−0.29	Sb–Ta	−0.51
P–Mn	0.12	Cr–Se	−0.03	Al–P	−0.15	Cr–Cr	−0.29	Au–U	−0.52
Zn–Pd	0.11	Sc–Ga	−0.03	Nb–Pt	−0.15	Al–W	−0.29	Ge–Np	−0.52
B–Te	0.11	Ru–Hf	−0.03	Sc–Cr	−0.15	Cr–Nb	−0.29	Hf–Hf	−0.52
Be–Al	0.11	P–Nb	−0.04	P–In	−0.15	Ge–Ta	−0.29	In–W	−0.52
Se–Pd	0.11	V–Ru	−0.04	Cr–Sn	−0.15	Se–W	−0.29	Hf–Ta	−0.52
Ni–Ga	0.11	Co–Pd	−0.04	Ru–U	−0.15	Ru–Pt	−0.29	Y–W	−0.52
Zn–In	0.10	Ti–Au	−0.04	Sb–Pt	−0.15	Cr–Ta	−0.29	Zr–Ta	−0.53
Zn–Au	0.10	Sc–Mn	−0.04	Se–Ir	−0.15	V–Y	−0.29	Mo–Np	−0.53
Be–Ga	0.10	V–Co	−0.04	Al–Ge	−0.15	Ge–Se	−0.30	Sb–W	−0.53

Be–Rh	0.09	B–V	−0.04	As–Nb	−0.15	As–In	−0.30	Nb–Np	−0.53
B–Ta	0.09	Sn–Au	−0.04	Ni–W	−0.15	Ge–Re	−0.30	Re–Au	−0.53
Cu–Pt	0.08	B–P	−0.04	Be–U	−0.15	Ga–Ta	−0.30	Te–Re	−0.53
Y–Rh	0.08	Co–Ni	−0.04	In–In	−0.15	V–Gd	−0.30	Os–Au	−0.53
Ge–Rh	0.08	Si–Sc	−0.04	S–Os	−0.15	Y–In	−0.30	Hf–Np	−0.54
Mn–Se	0.08	B–Co	−0.04	Ni–U	−0.15	Sn–Os	−0.31	Gd–W	−0.54
Ni–Zn	0.08	Si–Zn	−0.05	Si–Mo	−0.16	Y–Te	−0.31	Ge–U	−0.54
Ga–Au	0.08	Co–Au	−0.05	V–Ge	−0.16	Sc–Np	−0.31	Gd–Gd	−0.54
As–Rh	0.08	Cu–Ru	−0.05	Ni–Np	−0.16	Ga–Re	−0.31	Te–W	−0.55
S–Sn	0.08	Y–Ir	−0.05	Ni–Ru	−0.16	As–W	−0.31	Y–Ta	−0.55
Be–S	0.08	Nb–Rh	−0.05	Cu–Np	−0.16	V–Nb	−0.31	Y–U	−0.55
Ni–Te	0.08	Y–Ru	−0.05	Mn–Hf	−0.16	Al–Ta	−0.31	Al–Gd	−0.56
S–Ru	0.08	Cu–Nb	−0.05	V–Au	−0.16	V–Hf	−0.31	Np–U	−0.56
Sc–Ir	0.07	Be–Os	−0.05	In–Ir	−0.16	As–Re	−0.32	Np–Np	−0.57
Co–Y	0.07	In–Pt	−0.05	Zr–Au	−0.16	Si–Si	−0.32	Ge–Sb	−0.57
P–Ru	0.07	Al–Al	−0.05	Zn–Sb	−0.17	Ti–Gd	−0.32	In–Sb	−0.58
Sc–Co	0.07	Se–Au	−0.06	Si–Zr	−0.17	V–Zr	−0.32	Mo–U	−0.58
Ti–Rh	0.07	Co–Rh	−0.06	Cr–Sb	−0.17	Cr–W	−0.32	Zr–U	−0.59
Si–Mn	0.07	V–Se	−0.06	Cr–Os	−0.17	Ir–Ir	−0.32	Si–Te	−0.59
B–Rh	0.07	Sn–Pt	−0.06	Sc–Sn	−0.17	P–P	−0.32	Sn–Sn	−0.59
Mn–Cu	0.07	P–Zn	−0.06	Zn–Nb	−0.17	V–W	−0.32	Nb–U	−0.60
Rh–Hf	0.07	S–Ir	−0.06	Rh–Pd	−0.17	Pt–Np	−0.32	Ta–Np	−0.61
B–Ge	0.07	Mn–Au	−0.06	Cr–Te	−0.17	P–Ge	−0.32	Y–Gd	−0.62
Al–Zn	0.07	Ni–Nb	−0.06	Si–Nb	−0.18	V–Ta	−0.32	Hf–U	−0.62
Sc–As	0.06	Co–Pt	−0.06	As–Mo	−0.18	Ti–W	−0.33	As–Sb	−0.62
Al–Ir	0.06	Ni–Ta	−0.06	P–Se	−0.18	Ti–Nb	−0.33	Re–Np	−0.62
Be–Be	0.06	Mn–Te	−0.06	Cr–In	−0.18	Ge–W	−0.33	As–Te	−0.64
Pd–Sb	0.06	Mn–Ru	−0.06	Cu–U	−0.18	Sc–Nb	−0.33	U–U	−0.64
Mn–As	0.06	Ga–Ru	−0.06	Mn–Mo	−0.18	Pt–Pt	−0.33	Ge–Te	−0.64
Be–Ge	0.06	Zr–Pt	−0.06	Ge–Zr	−0.18	Ti–Y	−0.33	W–Np	−0.65
S–As	0.06	Ta–Ir	−0.06	Se–Ta	−0.18	In–Gd	−0.33	In–Np	−0.65
S–Te	0.06	Ge–Ir	−0.06	P–Np	−0.18	Te–Au	−0.33	Re–U	−0.67
Se–Rh	0.06	Zn–Ir	−0.06	Si–Hf	−0.18	Se–Re	−0.33	In–Te	−0.68
Si–Pt	0.06	Be–Ta	−0.07	V–Ga	−0.18	Si–Ge	−0.33	Ta–U	−0.69
Zn–Ga	0.06	Zn–Se	−0.07	Sb–Ir	−0.18	In–Os	−0.33	In–U	−0.71
Be–Ir	0.06	P–V	−0.07	Al–Se	−0.18	Pd–Re	−0.34	W–U	−0.71
Be–B	0.06	Zr–Ir	−0.07	Sc–Sb	−0.19	P–As	−0.34	Sn–Np	−0.72
Al–Mn	0.06	Ga–Ir	−0.07	Rh–Rh	−0.19	Sc–Zr	−0.34	Te–Np	−0.75
Se–Zr	0.06	B–W	−0.07	Al–V	−0.19	Si–Se	−0.34	Sb–Np	−0.78
Si–Ru	0.05	Gd–Au	−0.07	Ru–Rh	−0.19	Mo–Re	−0.34	Sn–U	−0.79
S–Cr	0.05	Co–W	−0.07	Rh–Re	−0.19	Y–Y	−0.34	Sn–Sb	−0.80
Rh–Sn	0.05	V–Cu	−0.07	S–Re	−0.19	Te–Gd	−0.34	Te–U	−0.82
Be–Zr	0.05	V–Pt	−0.07	Ti–Cr	−0.19	Ru–Au	−0.34	Sb–U	−0.85
P–Zr	0.05	Se–Se	−0.07	P–W	−0.20	Ga–W	−0.34	Gd–Ta	−0.85
Si–Au	0.05	Al–Ga	−0.07	Hf–Os	−0.20	Pt–U	−0.34	Sn–Te	−0.89
Ga–Rh	0.05	Ru–Ta	−0.07	Al–Ti	−0.20	Se–Sn	−0.34	Sb–Sb	−1.00
Ni–Hf	0.05	Zr–Ru	−0.07	Mn–Zr	−0.20	Zn–Np	−0.35	Sb–Te	−1.06
Be–Ni	0.05	Co–Mo	−0.07	Se–In	−0.20	Al–Sb	−0.35	Te–Te	−1.07
Be–Sb	0.05	Se–Nb	−0.07	Zn–Mo	−0.20	Ti–Zr	−0.35	Ru–Gd	−6.09
Ti–Ir	0.05	Ti–As	−0.07	P–Re	−0.20	Mo–Mo	−0.35	Gd–Ir	−6.14
Mn–Ge	0.05	Cu–Ta	−0.07	Ge–Hf	−0.20	Ge–Ge	−0.35	Mn–Gd	−6.17
Co–Sn	0.05	Si–Cr	−0.07	Ge–Os	−0.20	Mo–Sn	−0.35	Sc–Gd	−6.23
Rh–Gd	0.05	B–Os	−0.07	Ti–Te	−0.20	Si–Sn	−0.35	Gd–Np	−6.27
B–Mn	0.05	Cr–As	−0.07	Si–Re	−0.20	Si–P	−0.36	Gd–Os	−6.30
Be–Hf	0.05	Co–Np	−0.07	Sb–Au	−0.20	Ir–Pt	−0.36	Gd–U	−6.32
S–Ga	0.04	Cr–Ru	−0.07	Cr–Y	−0.21	Re–Re	−0.36	Mo–Gd	−6.39
Cu–Rh	0.04	Co–U	−0.07	Zn–Os	−0.21	Sb–Os	−0.36		
Ti–Pt	0.04	Mn–Ni	−0.07	Ru–Ir	−0.21	Ti–Hf	−0.36		
P–S	0.04	Al–Os	−0.07	Pd–Pt	−0.21	Re–Pt	−0.36		
B–As	0.04	As–Au	−0.08	Si–Ga	−0.21	Sc–Hf	−0.37		

S4.2. Benchmarking

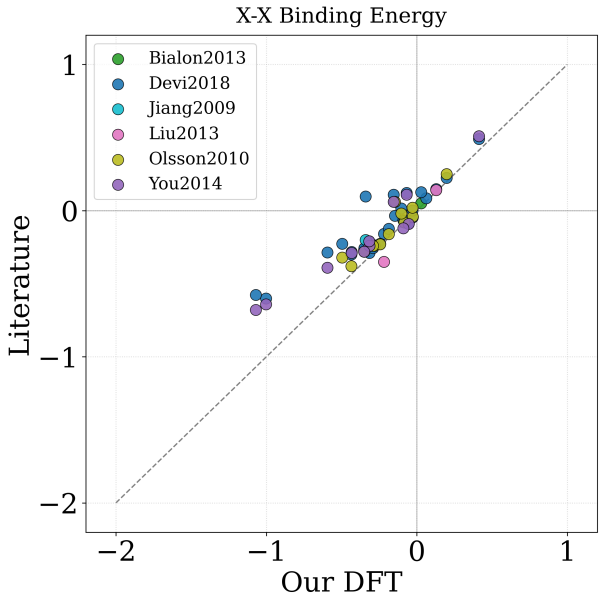


Figure S5: Comparison between our calculated and literature DFT values [13, 19, 20, 21, 22, 23] for the 1nn homoatomic solute pair binding energies (in eV) in bcc Fe.

S4.3. Analysis

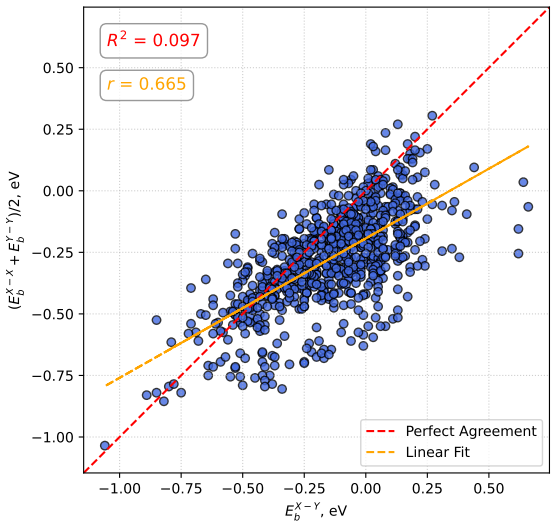


Figure S6: Correlation between the binding energy of a heteroatomic substitutional pair, (E_b^{X-Y}), and average value of two pure substitutional pairs, (($E_b^{X-X} + E_b^{Y-Y}$)/2) in bcc Fe. Red dashed line denotes perfect agreement between two values. The weak value (R^2=0.104) demonstrates the non-additive character of heteroatomic interactions. Orange dashed line denotes best linear fit with Pearson correlation coefficient (r=0.681) demonstrates near strong positive correlation between hetero- and average pure binding energies.

S5. Substitution-interstitial solutes interaction in bcc Fe

S5.1. Binding energies

Table S7 and Table S8 compile the calculated interaction energies between interstitial impurities and substitutional elements (X) in bcc Fe.

For the octahedral interstitials (C and N), Table S7 reveals a strong thermodynamic repulsion ($E_b < 0$) with the majority of transition metals and p-block elements in the first nearest-neighbor (1nn) position. Noticeable attraction ($E_b > 0$) is rare and found only for specific cases, such as C–B and N–B in the 1nn configuration.

As shown in Table S8, the tetrahedral interstitial hydrogen (H) exhibits significantly weaker interaction magnitudes compared to C and N. Early transition metals like Sc, Ti, Zr, and Hf demonstrate a weak attractive behavior ($E_b > 0$) towards hydrogen.

Table S7: Calculated binding energies E_b (in eV) for the 1nn and 2nn configurations between interstitial solute atoms (C, N) in octahedral positions and substitutional elements (X) in the bcc lattice of iron.

	Carbon (C-X)			**Nitrogen (N-X)**	
Element X	E_b (1nn), eV	E_b (2nn), eV	Element X	E_b (1nn), eV	E_b (2nn), eV
C–Be	0.166	−0.342	N–Be	–	–
C–B	0.618	−0.903	N–B	0.735	−1.214
C–Al	−0.693	−0.351	N–Al	−0.409	−0.299
C–Si	−0.592	−0.812	N–Si	−0.472	−0.782
C–P	−0.500	−0.996	N–P	−0.566	−0.737
C–S	−0.434	−0.526	N–S	−0.653	−0.189
C–Sc	−1.170	0.136	N–Sc	−0.661	0.420
C–Ti	−0.866	−0.008	N–Ti	−0.447	0.175
C–V	−0.590	−0.177	N–V	−0.246	−0.070
C–Cr	−0.363	−0.312	N–Cr	−0.131	−0.236
C–Mn	−0.055	−0.171	N–Mn	0.069	−0.101
C–Fe	0.000	0.000	N–Fe	0.000	0.000
C–Co	−0.145	−0.150	N–Co	−0.248	−0.211
C–Ni	−0.301	−0.249	N–Ni	−0.417	−0.315
C–Cu	−0.512	−0.214	N–Cu	−0.526	−0.222
C–Zn	−0.928	−0.470	N–Zn	−0.859	−0.408
C–Ga	−1.165	−0.663	N–Ga	−1.073	−0.572
C–Ge	−1.293	−0.898	N–Ge	−1.303	−0.787
C–As	−1.443	−1.062	N–As	–	−0.844
C–Se	−1.580	−0.905	N–Se	−1.686	–
C–Zr	−1.661	−0.044	N–Zr	–	0.157
C–Nb	−1.451	−0.146	N–Nb	−1.159	–
C–Mo	−1.143	−0.296	N–Mo	−0.976	−0.277
C–Ru	−0.655	−0.203	N–Ru	−0.798	−0.290
C–Rh	−0.823	−0.236	N–Rh	−1.086	−0.359
C–Pd	−1.199	−0.380	N–Pd	−1.456	−0.462
C–Sn	−2.210	−0.792	N–Sn	−2.169	−0.728
C–Sb	−2.421	−1.047	N–Sb	−2.455	−0.972
C–Te	−2.637	−1.221	N–Te	−2.741	–
C–Hf	−1.572	−0.007	N–Hf	−1.143	0.183
C–Ta	−1.425	−0.155	N–Ta	−1.104	−0.081
C–W	−1.242	−0.343	N–W	−1.066	−0.363
C–Re	−1.082	−0.510	N–Re	−1.115	−0.616
C–Os	−0.942	−0.477	N–Os	−1.156	−0.653
C–Ir	−1.011	−0.411	N–Ir	−1.369	−0.636
C–Pt	−1.386	−0.571	N–Pt	−1.817	−0.765
C–Au	−1.871	−0.736	N–Au	−2.173	−0.760
C–U	−1.612	−0.722	N–U	−1.258	−0.577
C–Np	−1.497	−0.734	N–Np	−1.165	−0.589

Table S8: Calculated binding energies E_b (in eV) for the 1nn and 2nn configurations between interstitial solute atom (H) in tetrahedral position and substitutional elements (X) in the bcc lattice of iron.

Element X	E_b (1nn), eV	E_b (2nn), eV	Element X	E_b (1nn), eV	E_b (2nn), eV
H–Be	−0.012	0.006	H–Zr	0.060	0.060
H–B	0.025	−0.041	H–Nb	0.001	0.000
H–Al	−0.230	0.011	H–Mo	−0.184	−0.061
H–Si	−0.499	−0.043	H–Ru	−0.190	−0.082
H–P	−0.599	−0.064	H–Rh	−0.171	−0.034
H–S	−0.274	−0.029	H–Pd	−0.202	−0.005
H–Sc	0.141	0.140	H–Sn	−0.001	−0.002
H–Ti	0.001	0.047	H–Sb	−0.042	−0.041
H–V	−0.121	−0.023	H–Te	−0.075	−0.075
H–Cr	−0.185	−0.059	H–Hf	0.064	0.064
H–Mn	−0.185	−0.073	H–Ta	−0.013	−0.013
H–Fe	0.000	0.000	H–W	−0.177	−0.079
H–Co	–	−0.008	H–Re	−0.331	−0.121
H–Ni	–	−0.014	H–Os	−0.366	–
H–Cu	−0.079	0.036	H–Ir	−0.294	−0.082
H–Zn	−0.279	0.032	H–Pt	−0.347	−0.052
H–Ga	−0.010	−0.010	H–Au	−0.028	−0.028
H–Ge	−0.039	−0.039	H–U	−0.143	−0.143
H–As	−0.511	−0.053	H–Np	−0.138	−0.138
H–Se	–	−0.051	-		

S6. Machine learning

Figures S7, S8, S9 depict the results of training machine learning models (XGBoost [24], Gradient Boosted Trees [25, 26, 27] and Random Forest [25], respectively) based on the binding energies between two substitutional solutes (excluding pairs with Y, Gd, In) in bcc Fe calculated using DFT method (for details see section S1 and Table S6).

List of used descriptors (symbol δ denotes the value of the corresponding quantity relative to the value of the same quantity for Fe.):

1. $\delta\rho_{sum}$: Sum of the densities of equilibrium phases of the soluted elements (g/cm^3);
2. $\delta\rho_{diff}$: Difference of the densities of equilibrium phases of the soluted elements (g/cm^3);
3. $\delta\chi_{sum}$: Sum of the electronegativites of the solute elements (Pauling's scale);
4. $\delta\chi_{diff}$: Difference of the electronegativites of the solute elements (Pauling's scale);
5. μ_{sum}: Sum of magnetic moments of soluted atoms in bcc Fe (μ_B);
6. μ_{diff}: Difference of magnetic moments of soluted atoms in bcc Fe (μ_B);
7. $\delta T_{m,sum}$: Sum of the melting points of equilibrium phases of the soluted elements (K);
8. $\delta T_{m,diff}$: Difference of the melting points of equilibrium phases of the soluted elements (K);
9. δZ_{sum}: Sum of the atomic numbers of the solute elements;
10. δZ_{diff}: Difference of the atomic numbers of the solute elements;
11. δR_{sum}: Sum of the atomic radii of the solute elements (pm);
12. δR_{diff}: Difference of the atomic radii of the solute elements (pm);
13. $\delta E^b_{v,1nn,sum}$: Sum of the binding energy of the solute elements with a monovacancy on first coordination sphere (eV);
14. $\delta E^b_{v,1nn,diff}$: Difference of the binding energy of the solute elements with a monovacancy on first coordination sphere (eV);
15. $\delta E^b_{v,2nn,sum}$: Sum of the binding energy of the solute elements with a monovacancy on second coordination sphere (eV);
16. $\delta E^b_{v,2nn,diff}$: Difference of the binding energy of the solute elements with a monovacancy on second coordination sphere (eV);
17. $\delta E^b_{v,3nn,sum}$: Sum of the binding energy of the solute elements with a monovacancy on third coordination sphere (eV);
18. $\delta E^b_{v,3nn,diff}$: Difference of the binding energy of the solute elements with a monovacancy on third coordination sphere (eV);
19. $\delta N_{val,sum}$: Sum of the valences of the solute elements;
20. $\delta N_{val,diff}$: Difference of the valences of the solute elements;

Densities, electronegativities and atomic radii (see in table S2) were taken from 'mendeleev' Python library. Magnetic moments and binding energies with monovacancies were obtained from DFT calculations in the present work. Valences of elements were taken from [28].

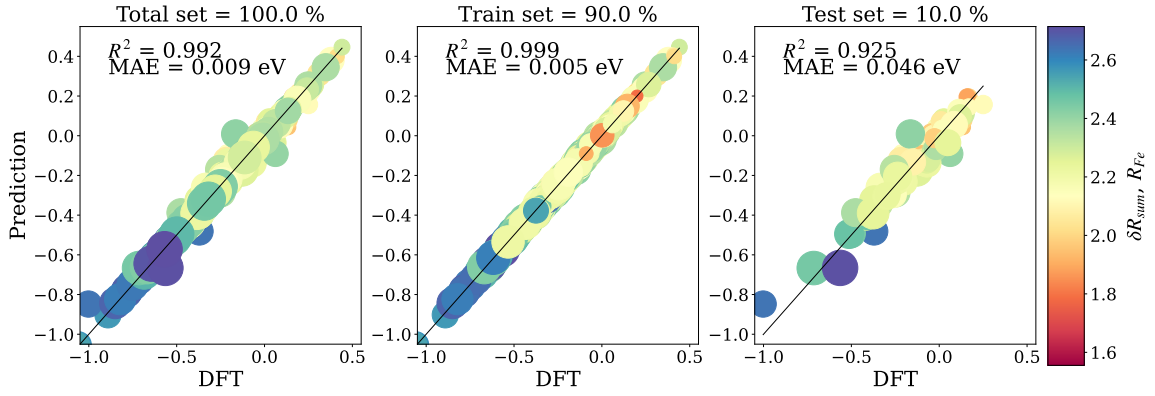


Figure S7: Scatter plot of machine-learning predictions for energies of binding between two substitutional solutes in the bcc Fe lattice versus DFT-PBE calculations. Machine-learning results are obtained using XGBoost regression model [24] with maximum depth 3 and number estimators 800. Sizes of symbols reflect the sum of atomic numbers of the two solutes. Colors reflect the sum of atomic radii relative to one of Fe. Units are eV.

Table S9 shows the feature importances of descriptors calculated using XGBoost model.

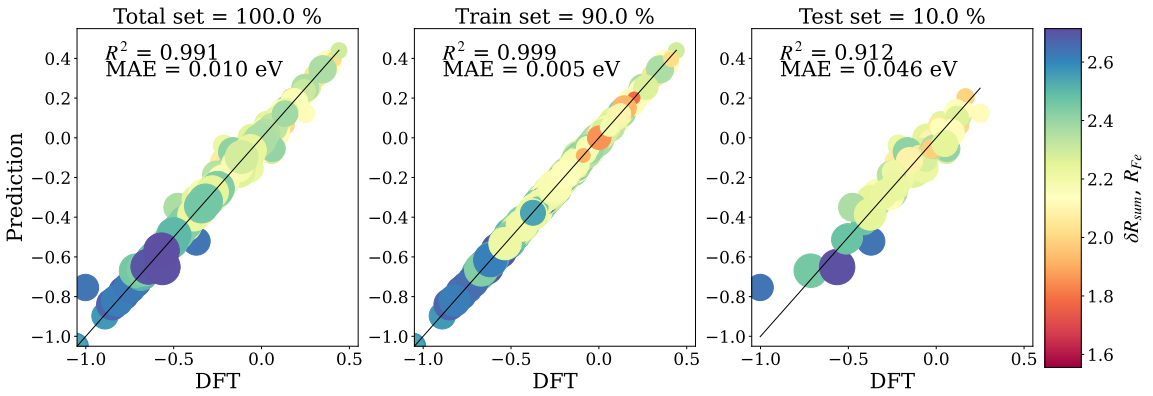


Figure S8: Scatter plot of machine-learning predictions for energies of binding between two substitutional solutes in the bcc Fe lattice versus DFT-PBE calculations. Machine-learning results are obtained using Gradient Boosted Trees regression model [25, 26, 27] with maximum depth 4 and number estimators 750. Sizes of symbols reflect the sum of atomic numbers of the two solutes. Colors reflect the sum of atomic radii relative to one of Fe. Units are eV.

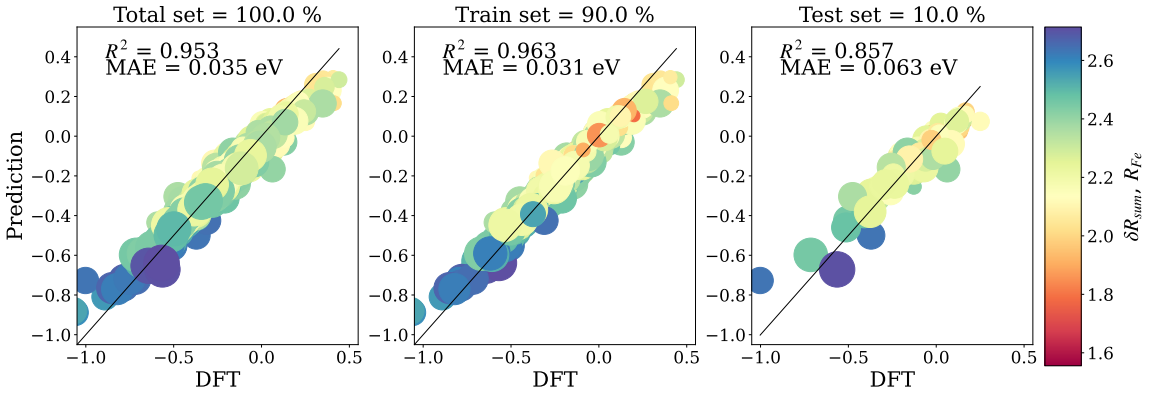


Figure S9: Scatter plot of machine-learning predictions for energies of binding between two substitutional solutes in the bcc Fe lattice versus DFT-PBE calculations. Machine-learning results are obtained using Random Forest regression model [25] with maximum depth ??? and number estimators 600. Sizes of symbols reflect the sum of atomic numbers of the two solutes. Colors reflect the sum of atomic radii relative to one of Fe. Units are eV.

Table S9: Feature importance scores of descriptors and their correlations with energy of binding between two substitutional solutes in the bcc Fe lattice, obtained using XGBoost regression model with maximum depth 3 and number estimators 800. ρ denotes density of a solute element (SE) in its equilibrium crystal phase, χ – Pauling's electronegativity of the SE atom, μ – magnetic moment of the SE atom, T_m – melting point of the SE in its equilibrium crystal phase, Z – atomic number of the SE, $E^b_{v,xnn}$ – energy of binding of the substitutional SE atom with vacancy on x-th coordination sphere, N_{val} – number of valence electrons in the SE atom, σ_b – difference between stresses in a sample with two solutes and the sum of stresses in samples with each solute separately, $E_{sub,max}$ – maximum replacement energy among the two solutes. Symbol δ denotes the value of the corresponding quantity relative to the value of the same quantity for Fe.

Descriptor	Importance, %	Correlation
$\delta\rho_{sum}$	1.075	–0.360
$\delta\rho_{diff}$	0.805	0.009
$\delta\chi_{sum}$	0.629	0.224
$\delta\chi_{diff}$	7.676	0.221
μ_{sum}	3.647	0.375
μ_{diff}	2.295	0.036
$\delta T_{m,sum}$	2.008	–0.157
$\delta T_{m,diff}$	1.354	–0.085
δZ_{sum}	14.065	–0.539
δZ_{diff}	1.994	–0.000
δR_{sum}	28.311	–0.604
δR_{diff}	5.226	0.202
$\delta E^b_{v,1nn,sum}$	2.649	–0.391
$\delta E^b_{v,1nn,diff}$	0.805	–0.171
$\delta E^b_{v,2nn,sum}$	3.715	0.301
$\delta E^b_{v,2nn,diff}$	0.720	0.076
$\delta E^b_{v,3nn,sum}$	0.565	–0.170
$\delta E^b_{v,3nn,diff}$	0.759	0.221
$\delta N_{val,sum}$	3.854	0.155
$\delta N_{val,diff}$	17.849	0.416

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